



ROB 2140 FINAL STUDY GUIDE

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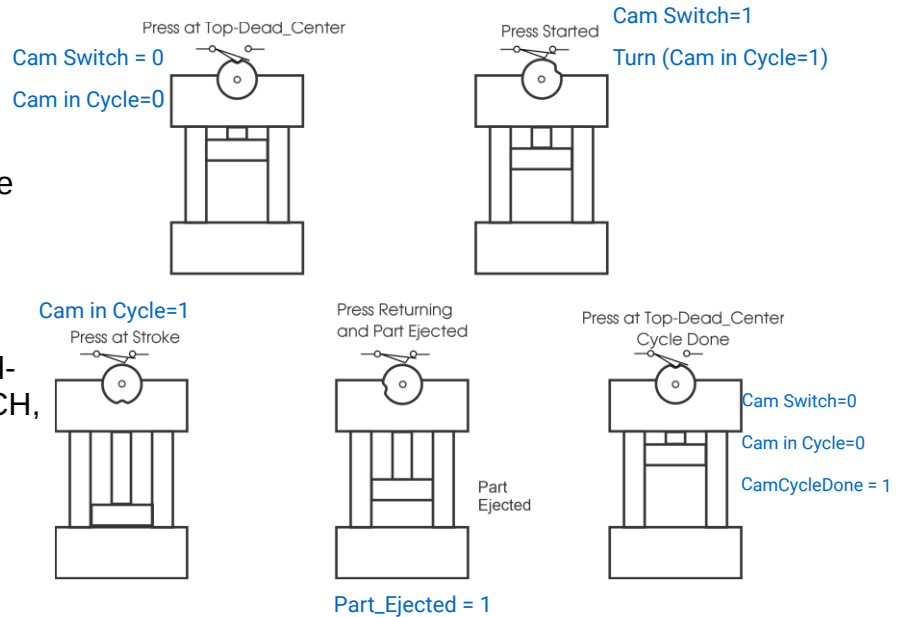
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SIEMENS CAM LOGIC EXAMPLES

The problem is similar to the BORE lab, “where the actions starts is the same condition where the action ends.”

The program monitors a normally open **CAM SWITCH** mounted on the cam to monitor a complete cycle of the cam turning to detect if a part is ejected from the press.

The cam-flywheel starts at top-dead-center and the switch, **CAM_SWITCH**, is in the normal position.



An **AUTO/MANUAL** switch is used to signal if the press machine is in operation to produce stamped part in **AUTO (Logic 1)** or in **MANUAL (Logic 0)** to reset any errors. Note: switching to Manual clears errors to allow for the machine to be moved if required to correct the error.

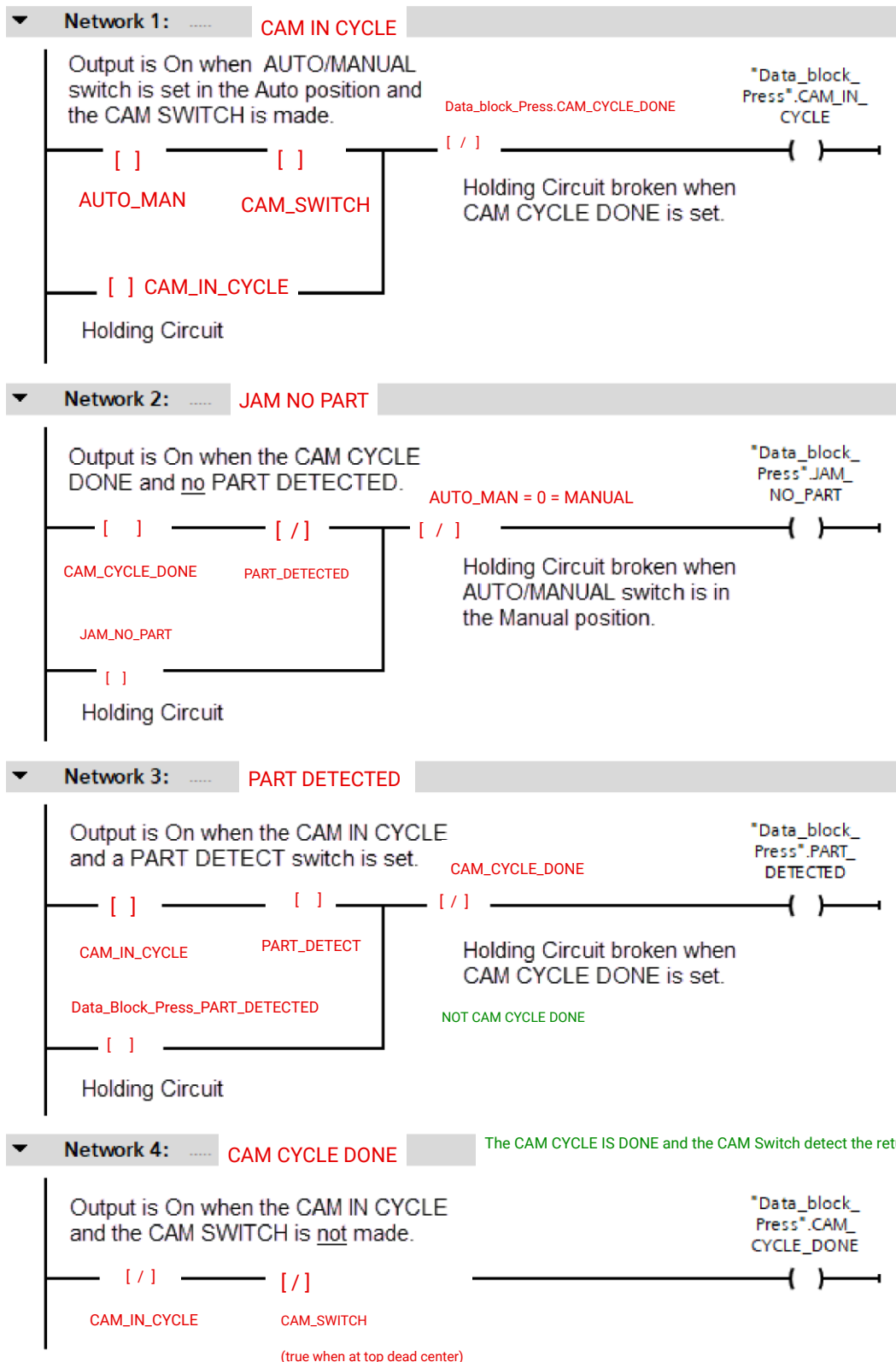
As the cam-flywheel begins to turn from top-dead-center a **CAM SWITCH** changes state. The change in state turns **ON** the **CAM IN CYCLE** output.

A switch monitors for a part to be ejected while the **CAM IS IN CYCLE** is **ON**. A part moving past the sensor turns on the **PART DETECTED** output while the part is in front of the sensor. The **PART DETECTED** output remains **ON** until the **CAM CYCLE DONE** output is turned **ON**.

If no part has been detected when the cam returns to top-dead-center a **JAM NO PART** error output is turned **ON** to stop the press ram motion. (See above, the error output is turned off by changing the Mode Switch to Manual.)

When the cam returns to top-dead-center the **CAM SWITCH** returns to the original state a **CAM CYCLE DONE** outputs turns **On** and the **CAM IN CYCLE** output turns OFF.

Equivalent Terms: Logic 1 = Set = Made



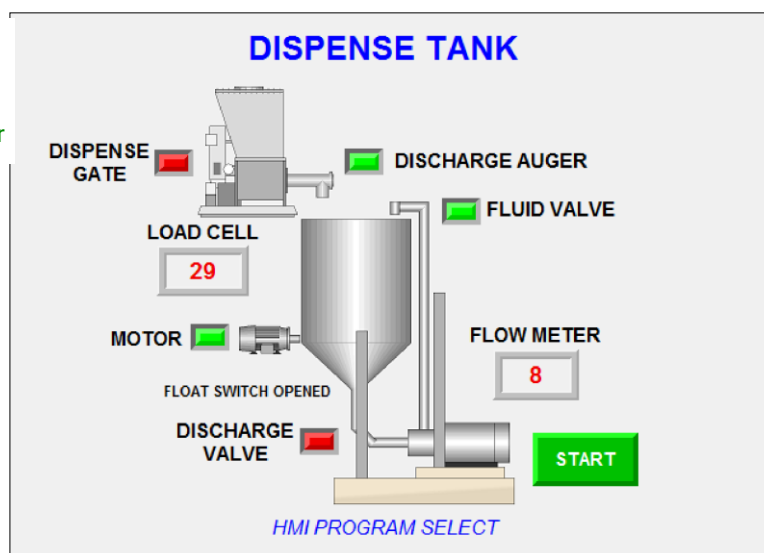
RSLOGIX 5000 ANALOG LOGIC EXAMPLES

NOTE: The final *will not use the* analog input modules to scale the value. The analog values are networked as raw bit values. An Add-On instruction will scale the sensor readings.

DO NOT USE LATCH -(L)- AND UNLATCH -(U)- INSTRUCTIONS. IF YOU DO NOT ACCOUNT FOR THE CONDITIONS TO CONTROL THE OUTPUTS CORRECTLY YOU WILL BE MARKED DOWN. USE OF LATCHES HAS BEEN A PROBLEM CONSISTENTLY FOR TEST TAKERS IN PAST.

Add on
To get Scaled Value for
Load Cell and Flow Meter

About 10 Rungs Total



Look at Add-On Instructions Hand
out

A solid material is dispensed onto a weighing platform when a **DISPENSE GATE** output is turned ON. When 35 pounds have been dispensed, the **DISPENSE GATE** closes by turning OFF the output.

The weight sensor signal that 35 pounds is on the **WEIGHTING PLATFORM** will trigger three output actions that will simultaneously occur:

- The **DISCHARGE AUGER** will turn ON to move the solid material into the vat.
- The output for the **DISPENSE LIQUID VALVE** will turn ON.
- The **MIXER MOTOR** will turn ON.

When the weight sensor at the weighing platform reads zero pounds, the **DISCHARGE AUGER** will turn OFF.

When the flow sensor signals that 70 gallons of liquid have been dispensed, the **DISPENSE LIQUID VALVE** will turn OFF.

Unless looking for exact value then don't use equal! Use a comparison may not ever be = !
we don't need limit in this lab just comparison

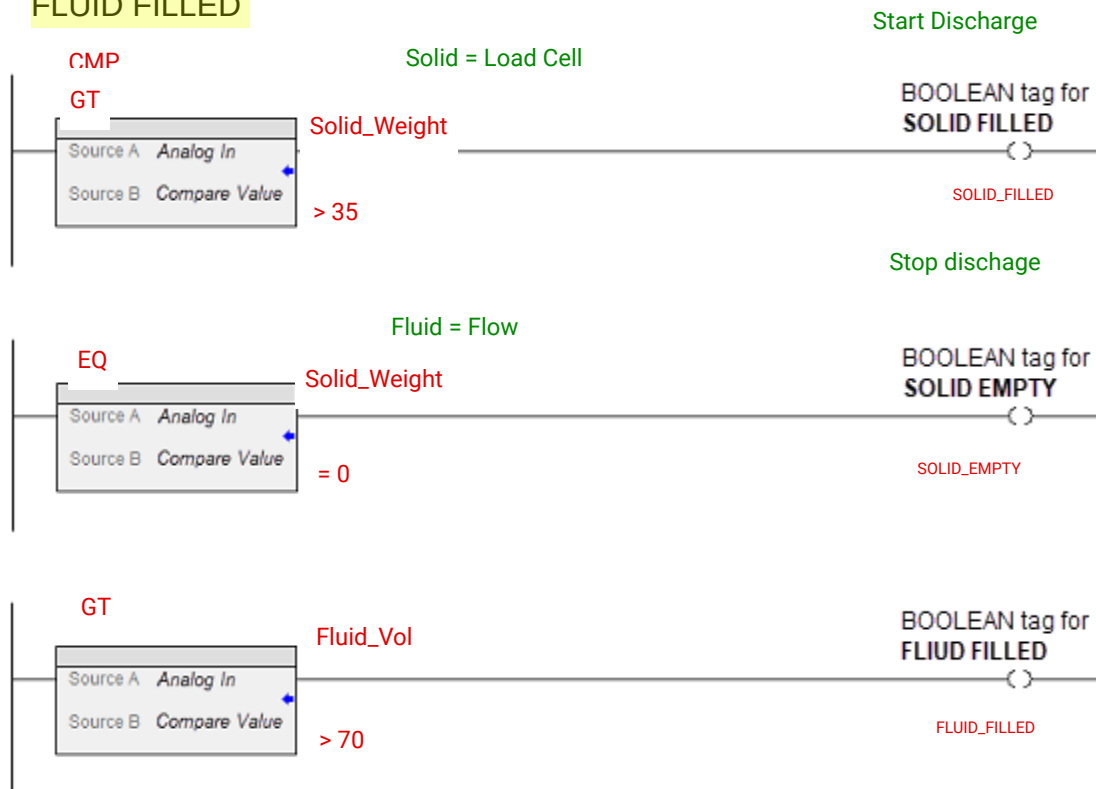
After a time delay the **MIXER MOTOR** will turn OFF.

When the MIX MOTOR TIMER times out, a DRAIN VALVE will turn ON. The DRAIN VALVE turns off when LOW LEVEL SWITCH closes. The level switch is wired Normally Closed but held Open by the mixture in the tank.

In this program, actions are taken when certain values are read from the Analog Inputs. By using Comparison Instructions, outputs tags to internal BOOLEAN Tags can be controlled when certain conditions (values) are read.

There are three BOOLEAN tags for the Measured Properties:

SOLID FILLED
SOLID EMPTY
FLUID FILLED



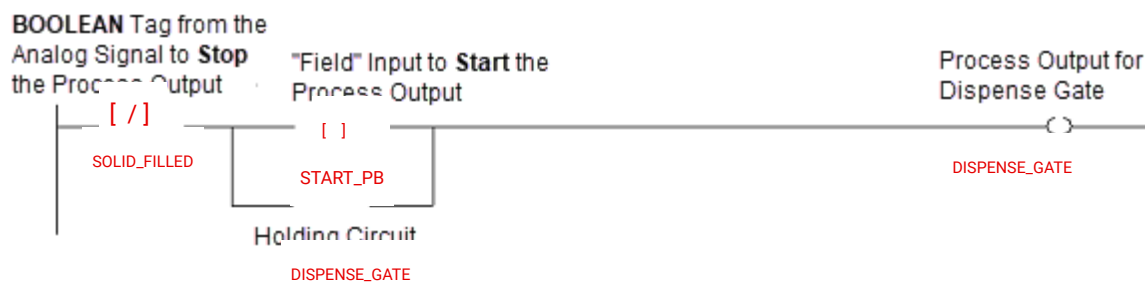
Do not use just an Equal To as the comparison instruction for Weighing Platform full and Fluid full binary addressed outputs. As discussed in the lectures, the rung would only be true if the value was exactly at that point when the rung was solved. It is better to use the Greater Than or Equal or Less Than or Equal comparison instructions, whichever is required by the logic, in case the value is a little below or above the point being determined.

The advantage of this programming approach is the “real world” Process Outputs are then controlled by holding circuits with either a Field “real world” Input or by the Internal BOOLEAN Tags, making or breaking the rung (like the holding “Start-Stop” circuit).

The Process Outputs being controlled are:

DISPENSE GATE	Start causes dispense Gate to turn on pushbutton so you need to hold it. When off? Stay on until you reached 35 pounds ideally
DISCHARGE AUGUR	What tells us when filled?
FLUID VALVE	Don't try three outputs same branch keep it simple!
MIXER MOTOR	Make three rungs three outputs to turn on.

The first “real world” output to control is the Solid Material DISPENSE GATE, which is turned ON, and held ON when the momentary START pushbutton is pressed and released. Solid fill turns on discharge auger off when no solid left ie empty



The output with a BOOLEAN tag for the SOLID FILLED, is used to “break” the input instruction for the holding circuit of the Solid Material Dispense Gate.

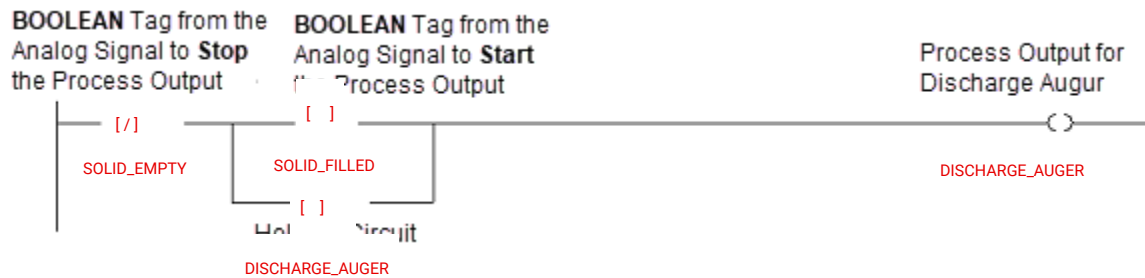
The SOLID FILLED Boolean tag is used to start the next three concurrent “real world” outputs.

NOTE: do not try and combine outputs onto a single rung, keep it simple by using individual rungs for the “real world” outputs.

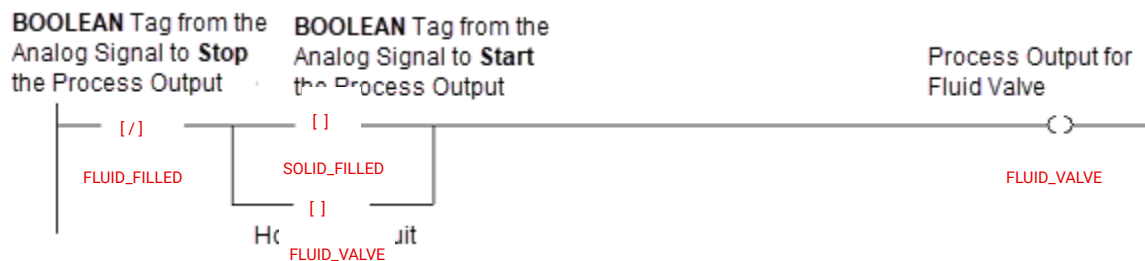
- Energize the output for the DISCHARGE AUGUR to move the solid material from the weighing platform to the tank.
- Turn On the output to open the FLUID VALVE.
- Turn On the output for the MIX MOTOR.

The signal which turns on the above outputs will be lost as the solid material is moved into the tank, but the outputs must be held ON.

Comparison up above tags are used below! Lots of start stop circuits

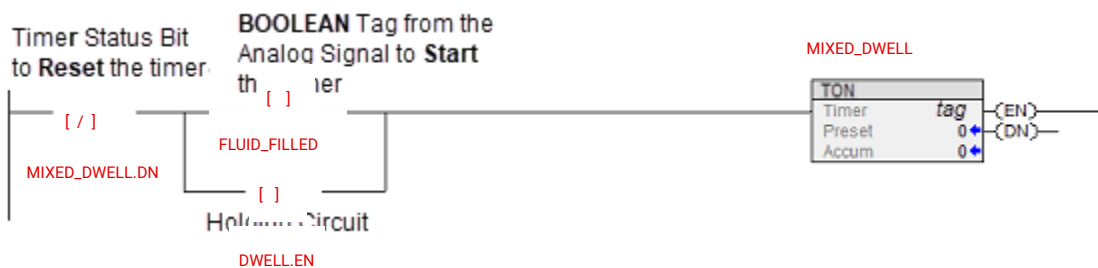


The BOOLEAN tag for the SOLID EMPTY, is used to “break” the holding circuit for the Solid Material Discharge Augur output.



To “break” the holding circuit to the Dispense Liquid Valve, the BOOLEAN tag for FLUID FILLED is used.

The BOOLEAN tag for **FLUID FILLED** starts the timer that will be used to turn ON the MIXER MOTOR.



The completed MIXER MOTOR timing cycle turns on the DISCHARGE VALVE.

The DISCHARGE VALVE is held on until the LOW LEVEL SWITCH Normally Closed Held Open by the **fluid mix closes** to “break” the holding circuit.

Fluid fill off when filled up the tank.

(No illustration of the rung is provided)

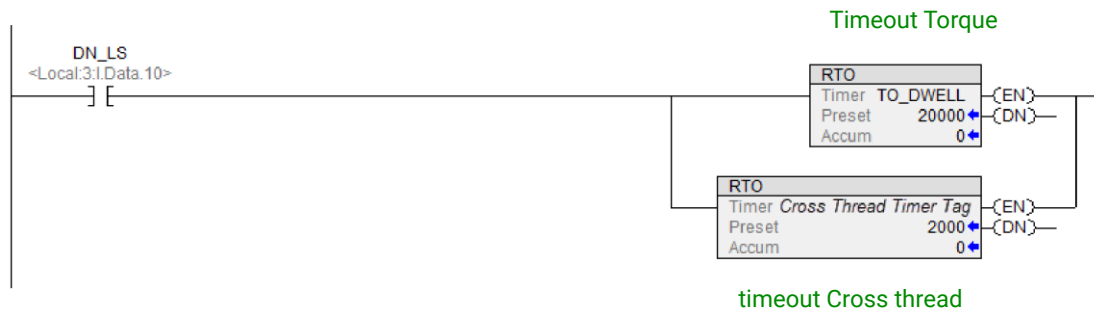
In Torque Lab we looked for Torque Error now we look for Cross Thread at beginning we see torque spike then it is cross thread. At beginning spike then cross thread. We need to add in cross thread. Same text as Torque Lab with cross threading added in. Any Tags specific for cross thread are in bold. Anything in bold is new. He will give you the torque lab also (without the cross thread).

RSLOGIX 5000 PROBLEM TORQUE LOGIC EXAMPLES

A second error condition for Cross Threading when the bolts are tightened is added.

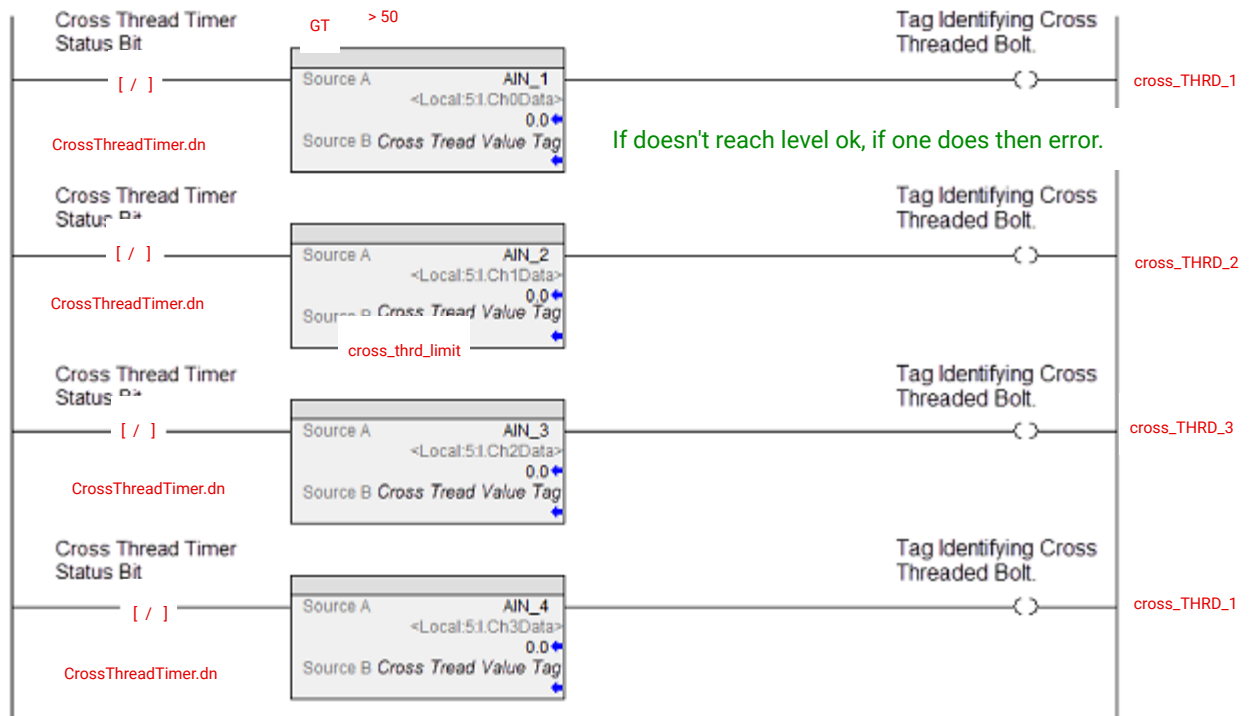
The **Torque Error** and **Cross Thread Error** will turn Off the nut runner motors and the **Down Solenoid** and **all Motors**. (*No illustration of the rungs is provided*)

A **Timer** is added for the Cross Thread Error, this timer will also begin elapsing when the down limit switch is set.



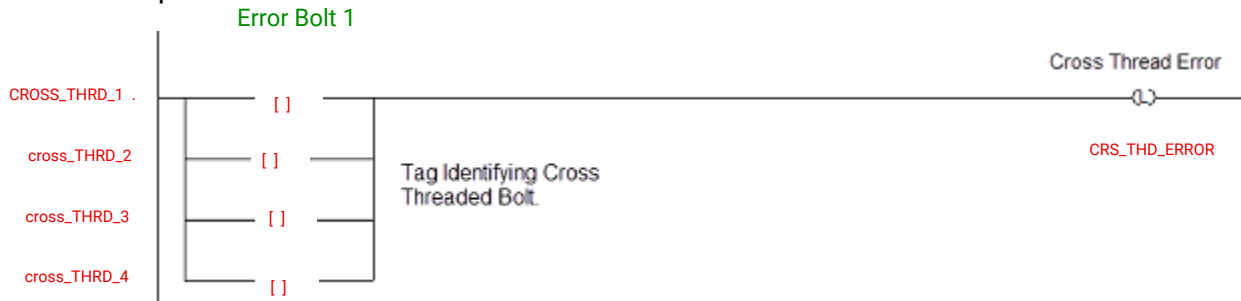
While the second timer is elapsing, a compare instruction will be monitoring each torque motor for setting a bit identifying the motor with the problem. Note: depending on the Timer Status bit used, the instruction may be the shown Examine Off -] / [- or the Examine On -] [-.

Looking at Analog sensor if reaches specific value too quickly Cross Thread Error on the bolts

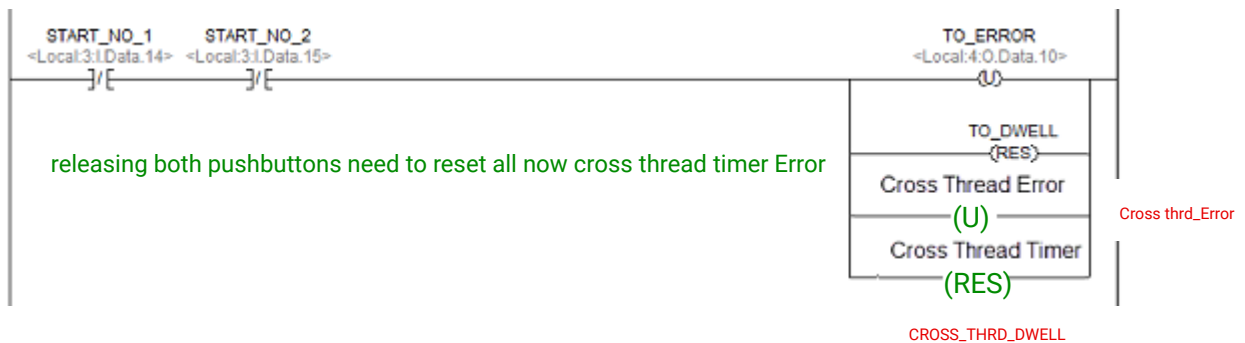


Latch on an error to indicate we have a cross thread. Monitor the bits below.

These Inputs identify the Cross Threaded Bolt bits that will latch the Cross Thread error output.

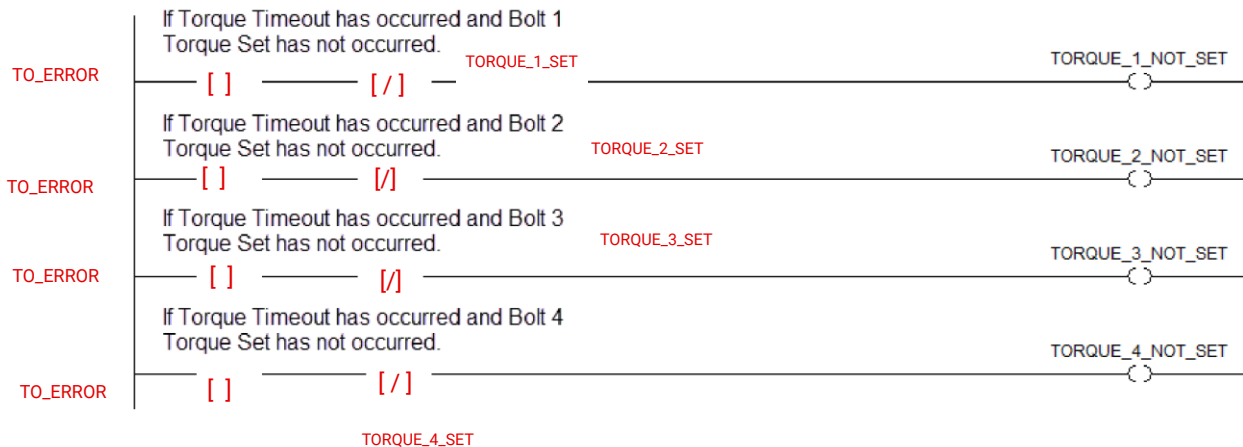


The releasing of both START pushbuttons will reset the timers and unlatch the error outputs.



Added for the HMI are outputs to indicate which bolt has not reached torque.

Original we didn't know which one indicated which one didn't make torque. Now we are adding outputs to show that.



TORQUE HMI

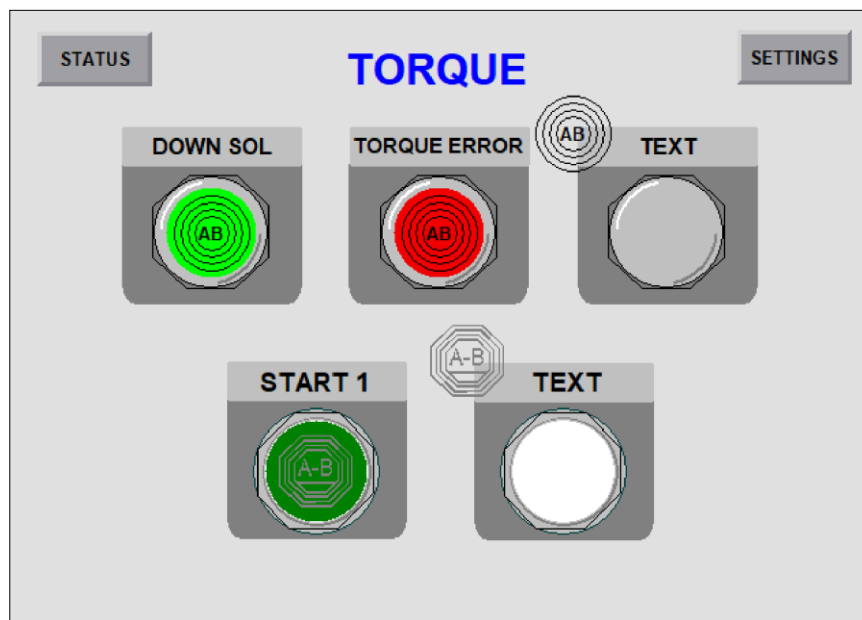
The HMI is from a template. Many of the objects are a repetitive process. Some of the repetitive objects will be completed. You will need to complete the objects that are undefined.

Object to be completed will be identified in the final.

FIRST PANEL

The first panel will be creating the Maintained Pushbutton and Multistate Indicator objects, the connections and arranging onto the graphic plates and embossed overlays.

The Navigation buttons for the go to screens will need to be defined.



This is the only panel where graphical objects are created. The second and third panels have the objects to be modified.

continued

recess is a separate graphic, graphic on top of a graphic object explorer

SECOND PANEL Set its properties etc. SLOW CLICKS !

The second panel has Numeric Displays which must be formatted for the proper number of digits and connections. What tab in the properties is for the value setting? The connections must be defined for each Numeric Display.



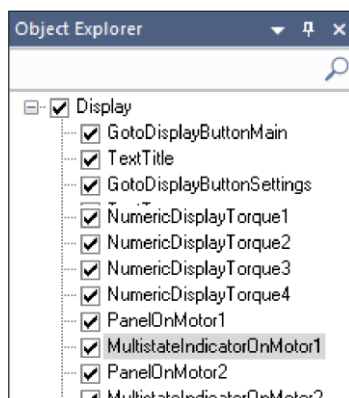
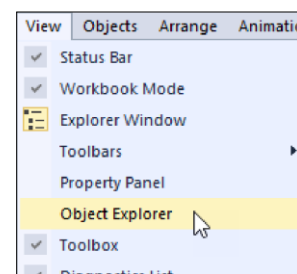
On the second panel are a twelve square Multistate Indicators. The indicator is inside a graphic panel object for the recessed effect.

The number of states, color, and the connection must be defined for each indicator.

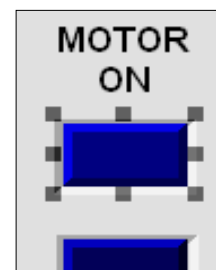
Method 1: Select MultiState Indicator by Object Explorer

On the Menu Bar, select **View** and then **Object Explorer**.

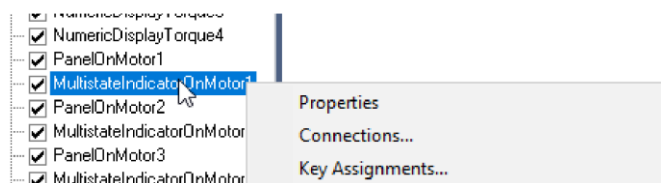
The Object Explorer displays the objects for the current display.



When an object is selected in the Object Explorer, the “drawing handles” appear around the object.



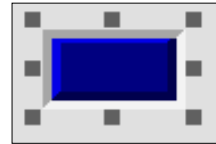
Right clicking on the selected object in the Object Explorer, will bring up a context menu and either **Properties** or **Connections...** may be selected.



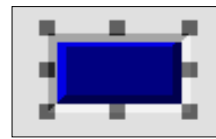
Method 2: Select MultiState Indicator by Single Left Click

DO NOT DOUBLE LEFT CLICK, THIS WILL OPEN THE PROPERTIES DIALOG ON THE OBJECT, MOST LIKELY THE RECESSED PANEL. THE METHOD BELOW IS A, SINGLE LEFT CLICK - PAUSE - SINGLE LEFT CLICK.

On the first single left click the recessed object is selected with its drawing handles.



Single left click again to select the Multistate Indicator. Notice the “drawing handles” are closer to the indicator.

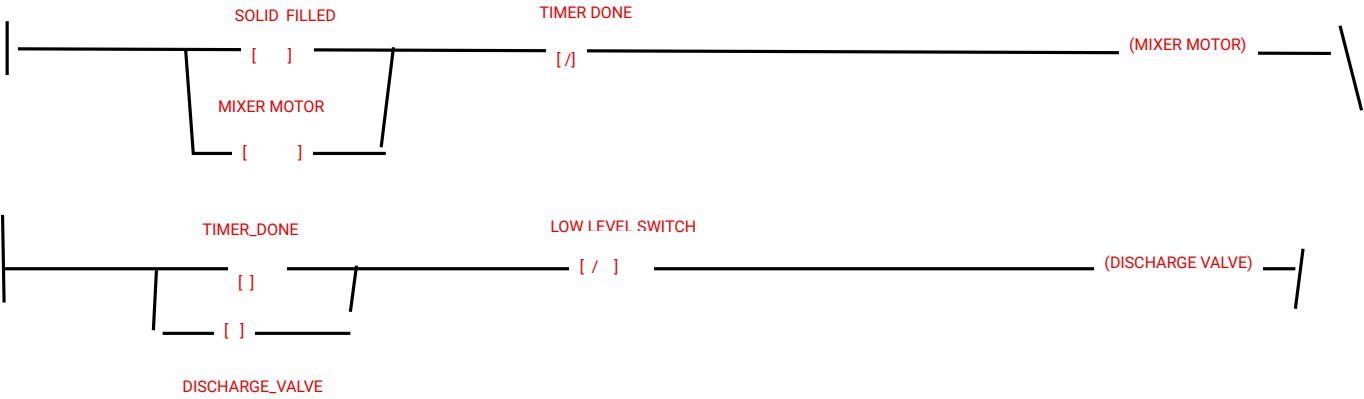


THIRD PANEL

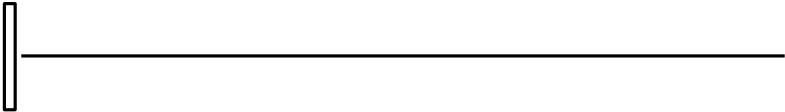
On the third panel, two Numeric Input Cursor Point objects must have the number format and connections defined.



Mixer Motor



These should be first



ADD ON INSTRUCTION

ANALOG_IN

SOLID

RAW_MAX

EU_MAX

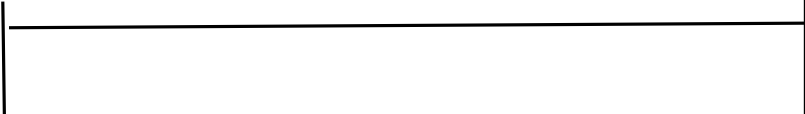
EU_MIN

ANALOG_CH

LOAD_CELL

EU_VA:I

SOLID_WEIGHT



ANALOG_IN

FLUID

RAW_MAX

EU_MAX

EU_MIN

ANALOG_CH

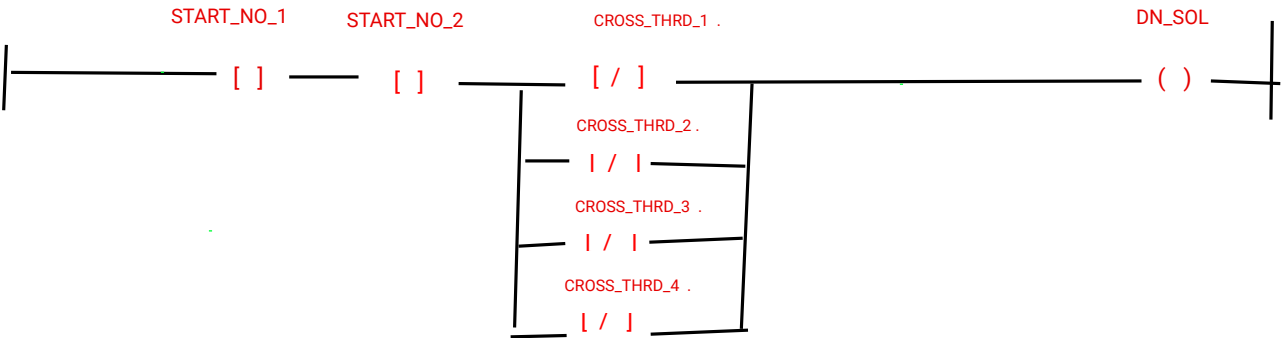
FLOW_METER

EU_VA:IE

FLUID_VOL

TORQUE

CROSS_THREAD_ERROR



TORQUE

CROSS_THREAD_ERROR

